# QUBO Formulation: Network Intrusion Feature Selection

## Network Intrusion Feature Selection

**Problem Statement.** Given a set of $N$ candidate network-traffic features, the goal is to select a subset of exactly $K$ features for an intrusion detection model. Each feature $i$ has an associated detection relevance score $s_i$, and each pair of features $(i, j)$ has an associated redundancy penalty $p_{ij}$. The objective is to maximize the total detection relevance of the selected features minus the total redundancy among the selected pairs. Formally, we seek a subset of indices of cardinality $K$ that maximizes the net score.

**Decision Variables.** Let $x_i \in \{0, 1\}$ be a binary decision variable for each candidate feature $i \in \{0, \ldots, N-1\}$, where $x_i = 1$ indicates that feature $i$ is selected, and $x_i = 0$ otherwise.

**QUBO Derivation.** We convert the maximization problem into a minimization problem suitable for a Quadratic Unconstrained Binary Optimization (QUBO) formulation by negating the objective function and incorporating the cardinality constraint as a penalty term.

**Step 1: Objective Function.** The original objective is to maximize $S(x) = \sum_{i=0}^{N-1} s_i x_i - \sum_{0 \le i < j \le N-1} p_{ij} x_i x_j$. Minimizing the negative of this objective yields the unconstrained QUBO component:

$$H_{\text{obj}}(x) = -\sum_{i=0}^{N-1} s_i x_i + \sum_{0 \le i < j \le N-1} p_{ij} x_i x_j. \tag{1}$$

**Step 2: Constraint.** The requirement to select exactly $K$ features is expressed as:

$$\sum_{i=0}^{N-1} x_i = K. \tag{2}$$

**Step 3: Penalty Term.** We introduce a penalty term with weight $\lambda > 0$ to enforce the

constraint. The penalty is proportional to the square of the constraint violation:

$$H_{\text{pen}}(x) = \lambda \left( \sum_{i=0}^{N-1} x_i - K \right)^2 . \tag{3}$$

Expanding the square and using the idempotent property $x_i^2 = x_i$ for binary variables:

$$\begin{aligned} \left( \sum_{i=0}^{N-1} x_i - K \right)^2 &= \sum_{i=0}^{N-1} x_i^2 + 2 \sum_{0 \le i < j \le N-1} x_i x_j - 2K \sum_{i=0}^{N-1} x_i + K^2 \\ &= \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \le i < j \le N-1} x_i x_j - 2K \sum_{i=0}^{N-1} x_i + K^2 \\ &= (1 - 2K) \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \le i < j \le N-1} x_i x_j + K^2. \end{aligned} \tag{4}$$

Thus, the penalty contribution is:

$$H_{\text{pen}}(x) = \lambda(1 - 2K) \sum_{i=0}^{N-1} x_i + 2\lambda \sum_{0 \le i < j \le N-1} x_i x_j + \lambda K^2. \tag{5}$$

**Step 4: Combined QUBO Hamiltonian.** Combining the objective and penalty terms, the total Hamiltonian $H(x) = H_{\text{obj}}(x) + H_{\text{pen}}(x)$ is:

$$H(x) = \sum_{i=0}^{N-1} \left[ -s_i + \lambda(1 - 2K) \right] x_i + \sum_{0 \le i < j \le N-1} \left[ p_{ij} + 2\lambda \right] x_i x_j + \lambda K^2. \tag{6}$$

**Step 5: Q Matrix and Offset.** The QUBO is defined by $H(x) = \sum_{i,j} Q_{i,j} x_i x_j + c$. Reading off the coefficients from the general form above, we obtain:

$$Q_{i,i} = -s_i + \lambda(1 - 2K), \quad \forall i \in \{0, \ldots, N-1\}, \tag{7}$$

$$Q_{i,j} = \frac{p_{ij}}{2} + \lambda, \quad \forall 0 \le i < j \le N-1, \tag{8}$$

$$c = \lambda K^2. \tag{9}$$

Here, $Q_{i,j}$ for $i \neq j$ represents the symmetric entry such that the term $x_i x_j$ in the quadratic form $\sum_{i,j} Q_{i,j} x_i x_j$ contributes $2Q_{i,j} x_i x_j$.

**Penalty Weight Justification.** The penalty weight $\lambda$ must be chosen sufficiently large to ensure that any violation of the cardinality constraint incurs a cost strictly greater than the maximum possible gain in the objective function. Let $\lambda > \max_i s_i$. The minimum penalty for a deviation of $\delta = |\sum x_i - K| \geq 1$ is $\lambda\delta^2 \geq \lambda$. The maximum gain in the objective function achievable by adding or removing features is bounded by the sum of relevance scores; however, for a deviation of $\delta$, the gain scales at most linearly with $\delta \cdot \max_i s_i$, while the penalty scales quadratically as $\lambda\delta^2$. Since $\lambda > \max_i s_i$, we have $\lambda\delta^2 > \delta \max_i s_i$ for all $\delta \geq 1$. Thus, any infeasible solution has an energy strictly higher than any feasible solution, ensuring the constraint is satisfied at the optimum.

**Correctness Argument.** Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal feature selection. For any feasible solution $x$ (where $\sum x_i = K$), the penalty term vanishes, and $H(x) = -S(x)$. Therefore, among feasible solutions, minimizing $H$ is equivalent to maximizing $S$. For any infeasible solution $x'$, the penalty term is positive. By the choice of $\lambda$, the penalty strictly dominates any potential increase in $S(x')$, implying $H(x') > H(x)$ for any feasible $x$. Consequently, the global minimum of $H(x)$ must occur at a feasible solution that maximizes the original objective.

**Illustrative Example.** Consider an instance with $N = 3$ candidate features and $K = 2$ features to be selected. The parameters are relevance scores $s_0, s_1, s_2$ and redundancy penalties $p_{01}, p_{02}, p_{12}$. Using the general formulas with $N = 3, K = 2$, and choosing $\lambda = 1000$ (satisfying $\lambda > \max_i s_i$), we instantiate the QUBO.

The penalty expansion coefficients are:

$$\lambda(1 - 2K) = 1000(1 - 4) = -3000, \tag{10}$$

$$2\lambda = 2000, \tag{11}$$

$$c = 1000 \cdot 2^2 = 4000. \tag{12}$$

The resulting Hamiltonian is:

$$\begin{aligned} H(x) = (-s_0 - 3000)x_0 + (-s_1 - 3000)x_1 + (-s_2 - 3000)x_2 \\ + (p_{01} + 2000)x_0x_1 + (p_{02} + 2000)x_0x_2 + (p_{12} + 2000)x_1x_2 + 4000. \end{aligned} \tag{13}$$

The feasible solutions are binary vectors with exactly two ones: $x^{(1)} = (1, 1, 0)$, $x^{(2)} = (1, 0, 1)$, and $x^{(3)} = (0, 1, 1)$. Evaluating $H$ on these yields:

$$H(x^{(1)}) = -s_0 - s_1 + p_{01}, \tag{14}$$

$$H(x^{(2)}) = -s_0 - s_2 + p_{02}, \tag{15}$$

$$H(x^{(3)}) = -s_1 - s_2 + p_{12}. \tag{16}$$

The optimal solution $x^*$ is the vector among $\{x^{(1)}, x^{(2)}, x^{(3)}\}$ that minimizes $H(x)$, which corresponds to the subset of two features maximizing $s_i + s_j - p_{ij}$. For instance, if $s = (10, 8, 6)$ and $p = (2, 3, 1)$, then $H(x^{(1)}) = -16$, $H(x^{(2)}) = -13$, $H(x^{(3)}) = -13$, so $x^* = (1, 1, 0)$ is optimal with energy $-16$.